

Exercise 1. Construct a bounded sequence in l^∞ that has no weakly convergent subsequence.

Exercise 2. Show that in a finite dimensional normed space weak convergence is equivalent to convergence in norm.

Exercise 3. Solve Exercise 2 on page 101, that is, show that in l^1 weak convergence is equivalent to convergence in norm.

Hint: Suppose we have a sequence (x^n) in l^1 with $\|x^n\| = 1$ which converges weakly to 0. Construct strictly increasing sequences of integers $(n_j), (p_j)$ such that

$$\sum_{k=1}^{p_{j-1}} |x_k^{n_j}| < \frac{1}{4}, \quad \sum_{k=p_{j-1}+1}^{p_j} |x_k^{n_j}| > \frac{3}{4}$$

and use this construction to derive a contradiction.